Inventory management and supply chain optimisation - DATA UNDERSTANDING

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FIELDS	DESCRIPTION
Category Id:	Product category code
Category Name:	Description of the product category
Customer City:	City where the customer made the purchase
Customer Country:	Country where the customer made the purchase
Customer Email:	Customer's email
Customer Fname:	Customer name
Customer Id:	Customer ID
Customer Lname:	Customer lastname
Customer Password:	Masked customer key
Customer Segment:	Types of Customers: Consumer , Corporate , Home Office
Customer State:	State to which the store where the purchase is registered belongs
Customer Street:	Street to which the store where the purchase is registered belongs
Customer Zipcode:	Customer Zipcode
Department Id:	Department code of store
Department Name:	Department name of store
Latitude:	Latitude corresponding to location of store
Longitude:	Longitude corresponding to location of store
Market:	Market to where the order is delivered : Africa , Europe , LATAM , Pacific USCA
Order City:	Destination city of the order
Order Country:	Destination country of the order
Order Customer Id:	Customer order code
order date (DateOrders):	Date on which the order is made
Order Id:	Order code
Order Item Cardprod Id:	Product code generated through the RFID reader
Order Item Discount:	Order item discount value
Order Item Discount Rate :	Order item discount percentage
Order Item Id:	Order item code
Order Item Product Price :	Price of products without discount
Order Item Profit Ratio:	Order Item Profit Ratio
Order Item Quantity:	Number of products per order
Sales:	Value in sales
Order Item Total :	Total amount per order
Order Profit Per Order:	Order Profit Per Order
Order Region:	Region of the world where the order is delivered : Southeast Asia ,South Asia ,Oceania ,Eastern Asia, West Asia , West of USA , US Center , West Africa, Central Africa ,North Africa ,Western Europe ,Northern , Caribbean , South America ,East Africa ,Southern Europe , East of USA ,Canada ,Southern Africa , Central Asia , Europe , Central America, Eastern Europe , South of USA
Order State:	State of the region where the order is delivered
Order Status:	Order Status : COMPLETE , PENDING , CLOSED ,

	PENDING_PAYMENT ,CANCELED,PROCESSING ,SUSPECTED_FRAUD ,ON_HOLD ,PAYMENT_REVIEW
Product Card Id:	Product code
Product Category Id:	Product category code
Product Description:	Product Description
Product Image:	Link of visit and purchase of the product
Product Name:	Product Name
Product Price:	Product Price
Product Status:	Status of the product stock :If it is 1 not available , 0 the product is available
Type:	Type of transaction made
Days for shipping (real) :	Actual shipping days of the purchased product
Days for shipment (scheduled):	Days of scheduled delivery of the purchased product
Benefit per order:	Earnings per order placed
Sales per customer:	Total sales per customer made per customer
Delivery Status:	Delivery status of orders: Advance shipping , Late delivery , Shipping Shipping on time
Late_delivery_risk	Categorical variable that indicates if sending is late (1), it is not late (0).
Shipping date (DateOrders)	Exact date and time of shipment
Shipping Mode	The following shipping modes are presented : Standard Class , First Class , Second Class , Same Day

DATA CLEANING

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FIELDS	STEPS/METHOD
Category Id:	76 categories/Drop
Category Name:	-
Customer City:	-
Customer Country:	IN SPANISH -- 'EE. UU.': 'United States',
Customer Email:	DROP
Customer Fname:	-
Customer Id:	Drop
Customer Lname:	-
Customer Password:	DROP
Customer Segment:	-
Customer State:	'PR': 'Puerto Rico','CA/96768/91732': 'California','NY': 'New 'Florida','MA': 'Massachusetts'
Customer Street:	-
Customer Zipcode:	Fill nulls from customer state
Department Id:	1-12
15. Department Name:	-
16. Latitude:	- Drop
17. Longitude:	- Drop
Market:	-
Order City:	-
Order Country:	Spanish to standard
Order Customer Id:	Customer ID/DROP
order date (DateOrders):	Date and time - needs formatting/ renamed to Order Date and sorted data acc to it in asc order
Order Id:	Drop
Order Item Cardprod Id:	Drop
Order Item Discount:	
Order Item Discount Rate :	
Order Item Id	Drop
Order Item Product Price :	unit price without discount
Order Item Profit Ratio:	
Order Item Quantity:	
Sales:	unit price * quantity (without discount)
Order Item Total :	Total amount per order with discount
Order Profit Per Order:	Profit Per Order
Order Region:	corrected/standard
Order State:	
Order Status:	
Order Zipcode:	DROP as almost null
Product Card Id:	Drop
Product Category Id:	Drop
Product Description:	DROP - ALL NULL
Product Image:	DROP
Product Name:	
Product Price:	Unit price
Product Status:	

Type :	CASH,DEBIT, PAYMENT,TRANSFER
Days for shipping (real) :	0-6
Days for shipment (scheduled):	0-4
Benefit per order:	DROP - same as profit per order
Sales per customer:	Total sales per customer made per customer (for each customer)
Delivery Status:	-
Late_delivery_risk	-
Shipping date	DROP same as order date
Shipping Mode:	-

FEATURE ENGINEERING

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1. Time-Based Features

Feature Name	Derived From	Method	Description	Business Importance
year	Order Date	Extract year using df['Order Date'].dt.year	Represents the calendar year of the order.	Helps track yearly demand trends and long-term growth.
month	Order Date	Extract month using df['Order Date'].dt.month	Represents the calendar month of the order.	Useful for detecting monthly seasonality (e.g., peak sales in December).
quarter	Order Date	Extract quarter using df['Order Date'].dt.quarter	Represents the fiscal quarter of the order.	Helps in quarterly forecasting, budgeting, and inventory planning.
lead_time_diff	Days for shipping (real) − Days for shipment (scheduled)	Simple subtraction	Measures how early or late the shipment happened compared to the planned date.	Positive values indicate delays, negative values indicate early deliveries. Helps in **delivery performance analysis**.

Why time features matter:

- Many products have **seasonal patterns** (e.g., winter jackets sell in Q4).
- These features help the model learn **when** demand increases or decreases.

2. Lag Features - calculated both global and regionwise

Feature Name	Derived From	Method	Description	Business Importance
lag_1	monthly_qty per SKU	.groupby(prod_col)['monthly_qty'].shift(1)	Quantity of the same SKU ordered **in the the previous month**.	Captures short-term demand patterns.
lag_3	monthly_qty per SKU	.shift(3)	Quantity of the SKU **3 months ago**.	Captures quarterly seasonality.
lag_6	monthly_qty per SKU	.shift(6)	Quantity of the SKU **6 months ago**.	Captures half-year patterns or delayed demand.
lag_12	monthly_qty per SKU	.shift(12)	Quantity of the SKU **12 months ago**.	Captures yearly seasonality for seasonal products.

Why lag features matter:

- Demand often **follows past patterns**.
- Lagged values give the model **memory** of previous demand.
- Essential for models like ARIMA, XGBoost, and deep learning time series forecasting.

3. Rolling Statistics Features

Feature Name	Derived From	Method	Description	Business Importance
rolling_mean_3	monthly_qty per SKU	.groupby(prod_col)['monthly_qty'].rolling(window=3).mean()	3-month moving average of SKU demand.	Smooths short-term fluctuations, reveals underlying trend.
rolling_std_3	monthly_qty per SKU	.rolling(window=3).std()	3-month rolling standard deviation of SKU demand.	Measures demand volatility — important for safety stock calculations.

Why rolling stats matter:

- Helps reduce **noise** in the data.
- Indicates **demand stability** or unpredictability.
- Useful for **stock buffer calculations** — high volatility may need higher safety stock.

Exploratory Data Analysis and Visualisation

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Per region top 3 products in class A
Order+sales per region
Per region lag 1 and lag 3
Per region rolling mean and std